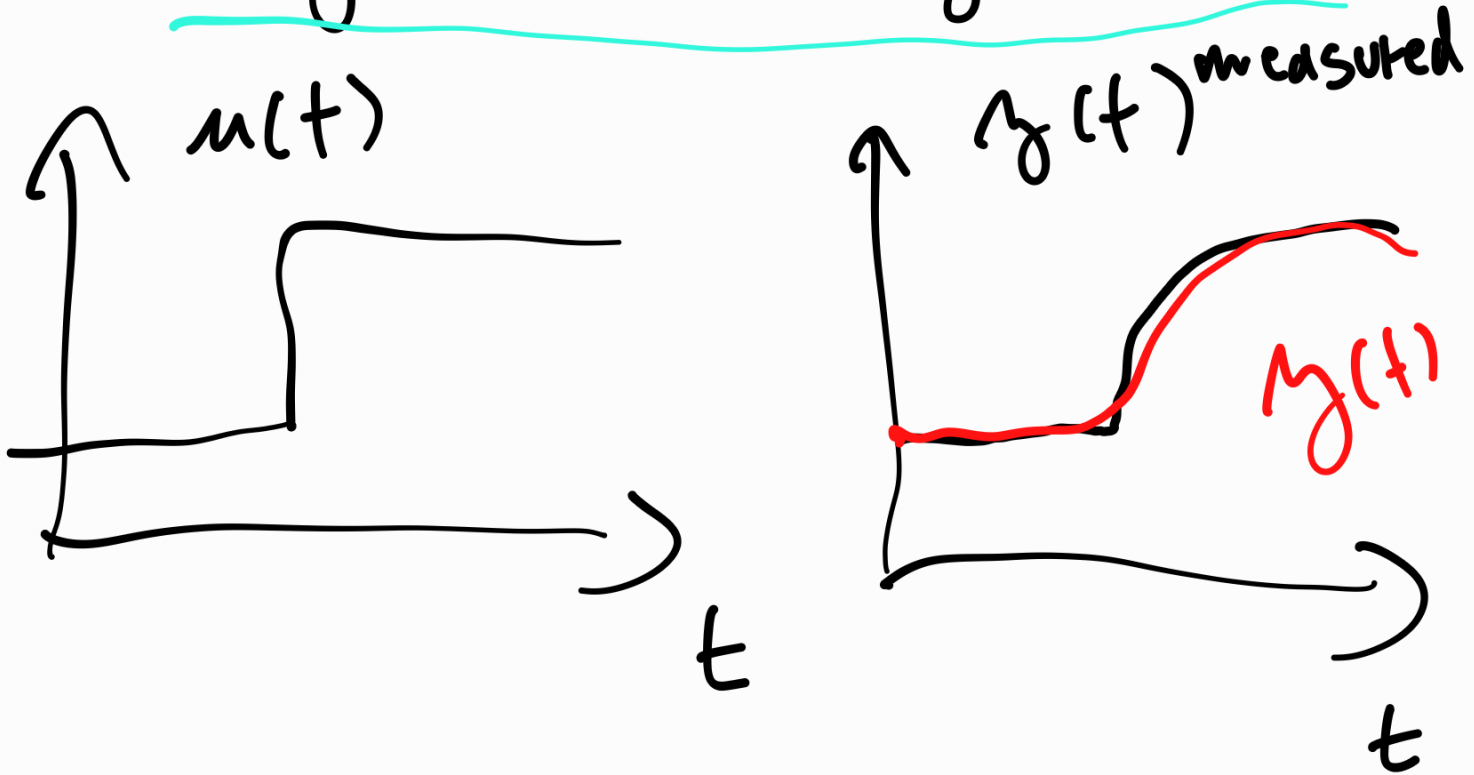
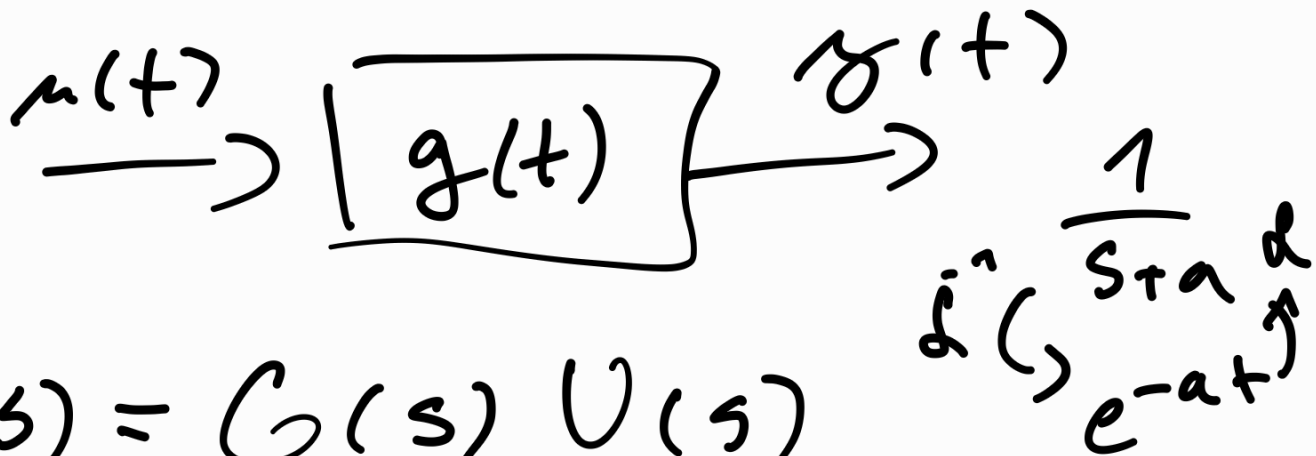


Identification of Dynamic Systems



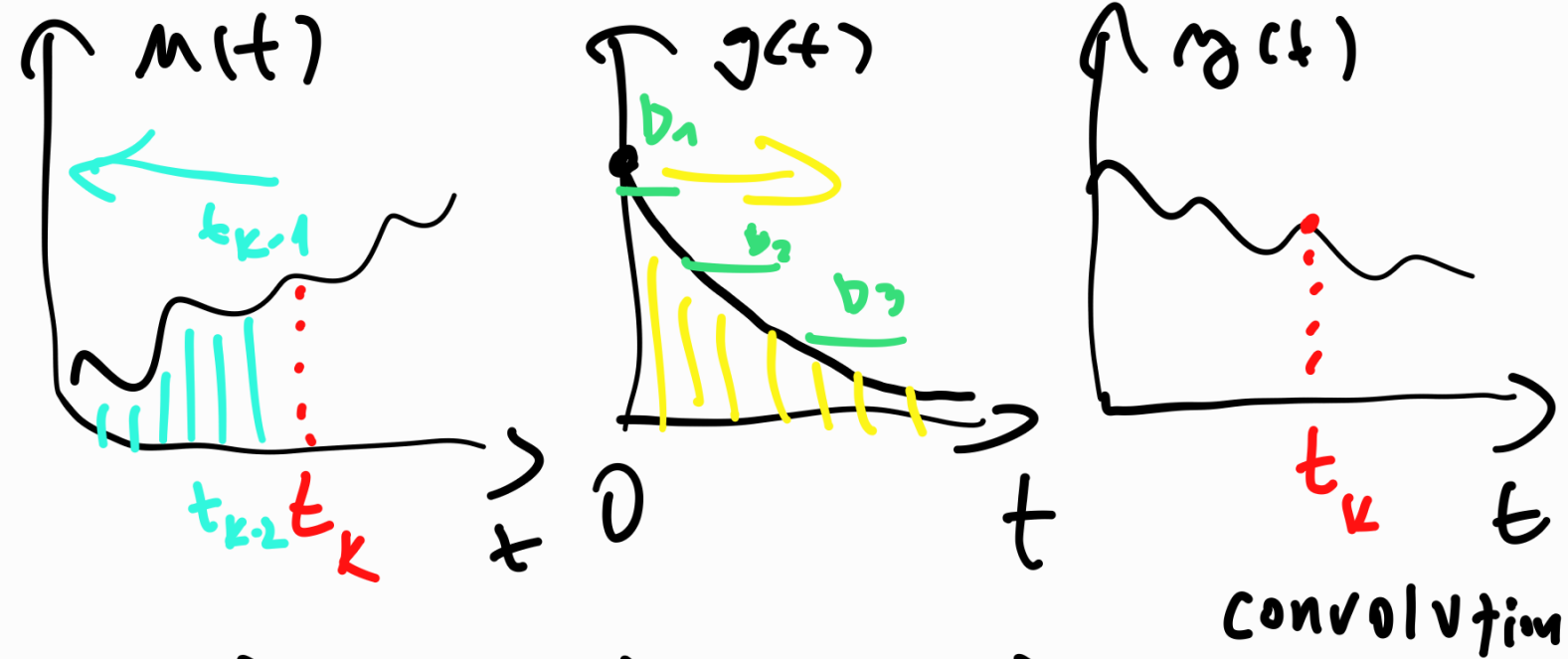
Finite Impulse Response



$$G(s) = \frac{1}{s + \frac{1}{T}} \rightarrow g(t) = e^{-\frac{t}{T}}$$

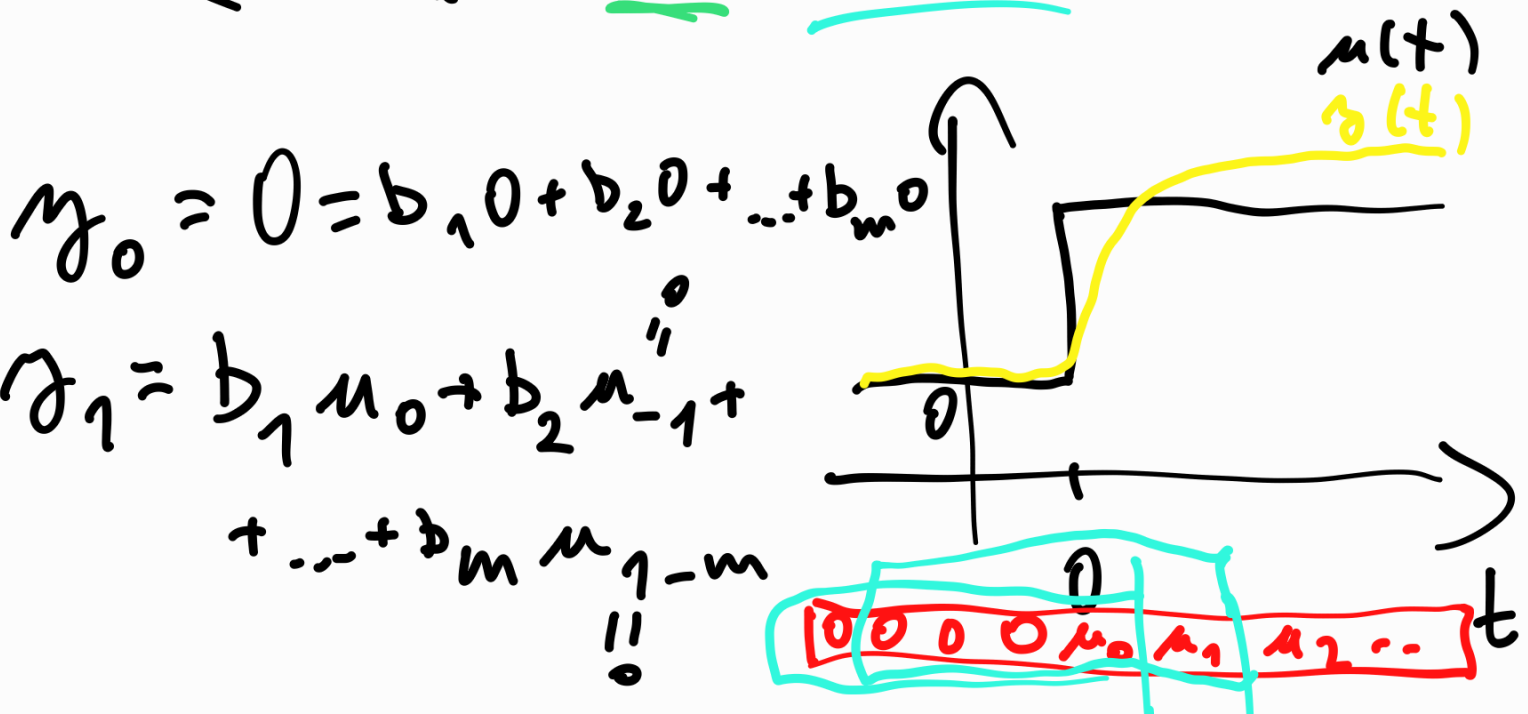
$u(t) = 1(t)$ step response

$u(t) = \delta(t)$ impulse response
 $V(s) = 1$ response



$$\begin{aligned}
 y(t) &= g(t) * u(t) = \\
 &= \int_0^t g(\tau) u(t-\tau) d\tau = \\
 &= \int_0^t \underline{g(t-\tau)} u(\tau) d\tau
 \end{aligned}$$

$$y_k = \sum_{i=1}^m \underline{b_i} \underline{u_{k-i}}$$



$$y_0 = 0 = b_1 0 + b_2 0 + \dots + b_m 0$$

$$\begin{aligned}
 y_1 &= b_1 u_0 + b_2 u_{-1} + \\
 &+ \dots + b_m u_{1-m}
 \end{aligned}$$

$$y_2 = b_1 \mu_1 + b_2 \mu_0 + \dots + b_m \mu_{2-m}$$

$\vdots$

$$y_m = b_1 \mu_{m-1} + b_2 \mu_{m-2} + \dots + b_m \mu_0$$

$\vdots$

$$y_N = b_1 \mu_{N-1} + b_2 \mu_{N-2} + \dots + b_m \mu_{N-m}$$

$$y = X P \quad P = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \\ \vdots \\ y_N \end{pmatrix} = \begin{pmatrix} \mu_0 & 0 & 0 & 0 & 0 & 0 \\ \mu_1 & \mu_0 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \mu_{m-1} & \mu_{m-2} & \dots & \mu_0 & & \\ \vdots & \vdots & & & & \\ \mu_{N-1} & \mu_{N-2} & \dots & \mu_{N-m} & & \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

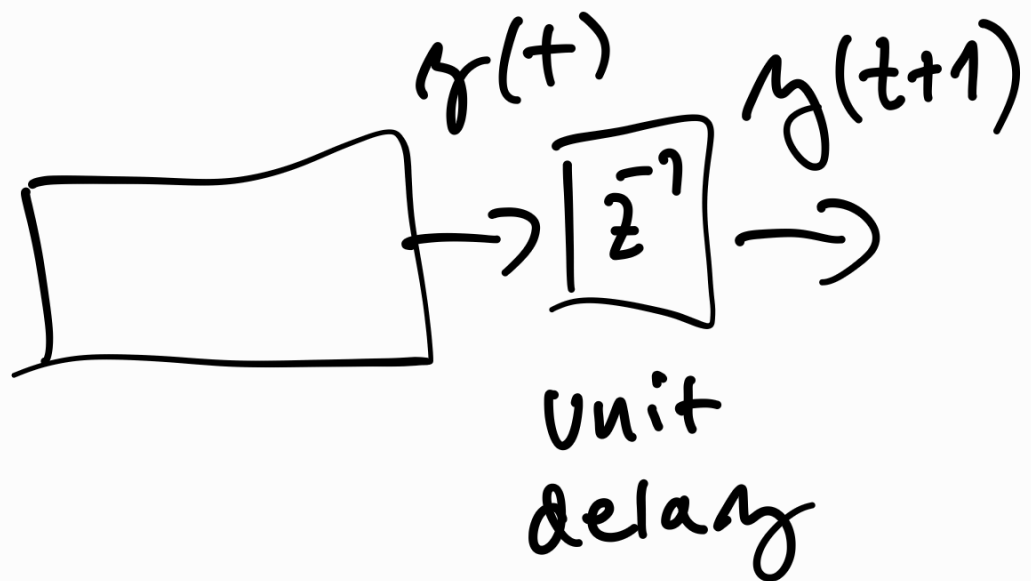
m - order of the model

$$N \geq m$$

AutoRegressive model with
exogenous input
 (ARX)

$$G(z^{-1}) = \frac{\sum_{i=1}^m b_i z^{-i}}{1 + \sum_{i=1}^n a_i z^{-i}} z^{-n_k}$$

input
delay



$$G(z^{-1}) = \frac{Y(z^{-1})}{U(z^{-1})} = \frac{\sum_{i=1}^m b_i z^{-i}}{1 + \sum_{i=1}^n a_i z^{-i}}$$

$$Y(z^{-1}) \left(1 + \sum_{i=1}^n a_i z^{-i} \right) =$$

$$U(z^{-1}) \sum_{i=1}^m b_i z^{-i} =$$

$$\underline{y}_{k-i} \longleftrightarrow z^{-i}$$

$$y_k = - \sum_{i=1}^n a_i y_{k-i} +$$

$$+ \sum_{i=1}^m b_i u_{k-i}$$

m/n order ARX model
 $n \geq m$

$$y_1 = -a_1 y_0 - a_2 y_{-1} - \dots - a_n y_{1-n} + b_1 u_0 + b_2 u_{-1} + \dots + b_m u_{1-m}$$

$$y_2 = -a_1 y_1 - a_2 y_0 - \dots - a_n y_{2-n} + b_1 u_1 + b_2 u_0 + \dots + b_m u_{2-m}$$

⋮

$$y_m = -a_1 y_{m-1} - a_2 y_{m-2} - \dots - a_n y_{m-n} + b_1 u_{m-1} + b_2 u_{m-2} + \dots + b_m u_0$$

$$y_n = -a_1 y_{n-1} - a_2 y_{n-2} - \dots - a_n y_0 + b_1 u_{n-1} + b_2 u_{n-2} + \dots + b_m u_{n-m}$$

$$- \dots - a_n y_0^+$$

$$+ b_1 u_{n-1} + b_2 u_{n-2}^+$$

$$+ \dots + b_m u_{n-m}$$

⋮

$$y_N = -a_1 y_{N-1} - a_2 y_{N-2}$$

$$- \dots - a_n y_{N-n}^+$$

$$+ b_1 u_{N-1} + b_2 u_{N-2}^+$$

$$+ \dots + b_m u_{N-m}$$

$$y = X P$$

$$P = \begin{pmatrix} a_1 \\ \vdots \\ a_n \\ b_1 \\ \vdots \\ b_m \end{pmatrix}$$

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \\ \vdots \\ y_n \\ \vdots \\ y_N \end{pmatrix} = \begin{pmatrix} -y_0 \ 0 \ 0 \ 0 & \mu_0 \ 0 \ 0 \ 0 \\ -y_1 \ -y_1 \ \dots & \mu_1 \ \mu_0 \ 0 \ 0 \\ -y_{m-1} \ -y_{m-2} \ -y_{m-n} & \mu_{m-1} \ \dots \ \mu_0 \\ -y_{n-1} \ \dots \ -y_0 & \mu_{n-1} \ \dots \ \mu_{n-m} \\ -y_{N-1} \ -y_{N-2} \ -y_{N-n} & \mu_{N-1} \ \dots \ \mu_{N-m} \end{pmatrix}$$